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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 13.8

$$f(x)=\sin\left(2x+\frac{\pi}{3}\right)$$

$$\text{Periode}=\pi$$

$$f^{-1}(0)=\left\{-\frac{\pi}{6},\frac{\pi}{3},\frac{5\pi}{6}\right\}$$

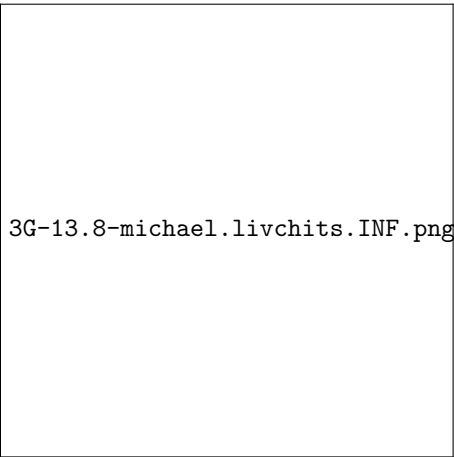
$$f'(x)=2\cos\left(2x+\frac{\pi}{3}\right)$$

$$(f')^{-1}(0)=\left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$$

$$f''(x)=-4\sin\left(2x+\frac{\pi}{3}\right)$$

$$(f'')^{-1}(0)=\left\{-\frac{\pi}{6},\frac{\pi}{3},\frac{5\pi}{6}\right\}$$

x	$-\frac{\pi}{6}$		$\frac{\pi}{12}$		$\frac{\pi}{3}$		$\frac{7\pi}{12}$		$\frac{5\pi}{6}$
$f(x)$	0	+	+	+	0	−	−	−	0
$f'(x)$	+	+	0	−	−	−	0	+	+
$f''(x)$	0	−	−	−	0	+	+	+	0
$f(x)$	$\nearrow$ $B(0)$	$\nearrow$ $\frown$	$M(1)$ $\frown$	$\searrow$ $\frown$	$\searrow$ $B(0)$	$\searrow$ $\smile$	$m(-1)$ $\smile$	$\nearrow$ $\smile$	$\nearrow$ $B(0)$



References